Coding Challenge: Customer Orders Analysis

Problem Statement

You are working as a **Data Analyst** for an e-commerce company.
Your task is to analyze customer purchases, orders, and payments using SQL JOIN operations.

You are given **3 tables**:

- customers
- orders
- payments

Write SQL queries to answer the business questions below.

Database Schema

customers

Column Name	Data Type
customer_id	INT (PK)
customer_name	VARCHAR
city	VARCHAR

orders

Column Name	Data Type
order_id	INT (PK)
customer_id	INT
order_date	DATE

Column Name	Data Type
amount	DECIMAL

payments

Column Name	Data Type
payment_id	INT (PK)
order_id	INT
payment_status	VARCHAR

Sample Data

```
CREATE TABLE customers (
   customer_id INT PRIMARY KEY,
   customer_name VARCHAR(100),
   city VARCHAR(50)
);

INSERT INTO customers VALUES
(1, 'Amit', 'Bangalore'),
(2, 'Sneha', 'Mumbai'),
(3, 'Rahul', 'Delhi'),
(4, 'Priya', 'Chennai');


CREATE TABLE orders (
   order_id INT PRIMARY KEY,
   customer_id INT,
   order_date DATE,
```

```
    amount DECIMAL(10,2)
);

INSERT INTO orders VALUES
(101, 1, '2024-01-10', 500),
(102, 1, '2024-02-15', 700),
(103, 2, '2024-03-01', 300),
(104, 5, '2024-03-05', 900);


CREATE TABLE payments (
    payment_id INT PRIMARY KEY,
    order_id INT,
    payment_status VARCHAR(20)
);

INSERT INTO payments VALUES
(1, 101, 'Completed'),
(2, 102, 'Pending'),
(3, 103, 'Completed');
```

Tasks

Task 1: Customer Orders

Write a query to display:

- customer_name
- order_id
- amount

Include only customers who placed orders.

Task 2: All Customers

Write a query to display:

- all customers
- their order_id (if any)

Customers without orders should still appear.

Task 3: Invalid Orders

Write a query to find:

- orders that do NOT have a matching customer

Task 4: Order Payment Status

Write a query to display:

- customer_name
- order_id
- payment_status

Include all orders, even if payment is missing.

Task 5: Customers Without Orders

Find customers who have **never placed an order**.

Task 6: Orders Without Payment

Find all orders that **do not have a payment record**.

Task 7: Total Spending

Write a query to calculate:

- total amount spent by each customer

Task 8: Fully Paid Customers

Find customers whose **all orders are marked as 'Completed'**.

Task 9: Highest Order Per Customer

Display:

- customer_name
- highest order amount

Task 10: Top Customers

Find **top 2 customers based on total spending**.

Rules

- Use appropriate JOIN types:
    - INNER JOIN
    - LEFT JOIN
    - RIGHT JOIN
- Avoid subqueries unless necessary
- Use GROUP BY where required
- Write clean and readable SQL